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Wide Vision Drilling (WVD) a Proven Roadmap for Mastering Complex Wells

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Abstract

The paper outlines the work process and well construction development for the complex Kvitebjørn and Valemon HPHT (High Pressure and High Temperature) fields in North Sea. It addresses the technical challenges of drilling a 4-kilometer, interbedded, near-horizontal, unstable 17 ½-inch section, which involved multiple trips due to failures and service quality incidents, often requiring sidetracks and re-drilling. Ultimately, this contributed to delivering wells 23-54% over budget. To overcome those challenges, it was crucial to develop methodology to ensure a sustainable approach that remains despite changes of project team.

The Wide Vision Drilling (WVD) is a roadmap that was developed based on thorough offset analysis technique and progressive learning strategy for wells drilled in Kvitebjørn and Valemon HPHT fields since 2004. It details all subsurface risk and breaks it down into risk packages relevant to different formation groups and develops a drilling strategy/roadmap for each group. It provides a detailed and systematic approach to ensure continuity of project risks understanding and field specific best practices implementation. This roadmap ensures safe and efficient section delivery, proving that optimal performance balances speed and risk control.

The WVD strategy led to a groundbreaking achievement on the Kvitebjørn and Valemon field, successfully delivering single run in every section drilled for the first time. This significantly increased delivery efficiency by 50-80% compared to project's historical average. The improved performance was achieved with zero HSE incidents and no well control issues. The project also realized cost savings, drilling wells with more than a 20- 30% reduction in budget time and reducing non-productive time (NPT) from 46% to 11%. Additionally, early production was accelerated by over 50 days and wells were delivered with all required well integrity barriers in place. This process marked a significant shift, transforming the field strategy and enabling Equinor to develop previously unfeasible targets, thereby extending the project's life.

These results have elevated Kvitebjørn and Valemon from challenging projects to world-class HPHT operation. Beyond these tangible outcomes, the initiative fostered a culture of continuous learning and performance within the One Team. The process's agility ensures that changes in personnel will not disrupt the project performance, maintaining consistent results. The WVD roadmap played a crucial role in improving the operational success of both fields.

Frequent personnel changes pose significant challenges in the Energy industry and hence field best practices are changed as newcomers come with their preferred practices. It is common that they start implementing those new practices without understanding the historical evolution of the fields design and best practices. This paper outlines a strategic process for developing unique local field practices, and mitigating negative impact from personnel changes

Introduction

Modern well construction is often approached through a technology-first lens. When challenges arise, the typical industry response is to add new tools, additional sensors, advanced bottom-hole assemblies, improved drilling fluids, or the latest bit designs. While these advancements are valuable, over-reliance on technology can conceal the core objective: understanding and managing the root causes of risk, rather than simply reacting to its symptoms.

In the complex HPHT environments of Kvitebjørn and Valemon fields in the North Sea, a different philosophy emerged. Drilling long, interbedded, near horizontal, and geomechanically unstable sections demanded more than incremental hardware upgrades. The team behind the Wide Vision Drilling (WVD) strategy recognized that sustainable performance would come not from additional complexity in tools, but from a deeper comprehension of geology, risk progression, and engineering context.

The WVD roadmap is more than a technical process, it is a disciplined mindset. It places understanding before intervention, ensuring that technology is applied only after risks are clearly defined, decomposed, and mitigated through engineering design. By integrating local field experience, detailed offset analysis, and continuous learning, the WVD approach delivered stable, efficient, and low-NPT performance in one of the most demanding HPHT operating environments in the world.

Kvitebjørn Field

Kvitebjørn is a gas and condensate field in the in the Tampen area in the northern part of the North Sea within the Norwegian sector, as shown in (Figure 1). Kvitebjørn produces from Middle Jurassic sandstone in the Brent Group with secondary reservoirs on the Lower Jurassic Cook formation and Upper Triassic Statfjord Group. The main reservoirs are encountered at approximately 4,000 mTVD-RKB, with initial HPHT conditions of 775 bar (11,240 psi) and 162 °C (324 °F).

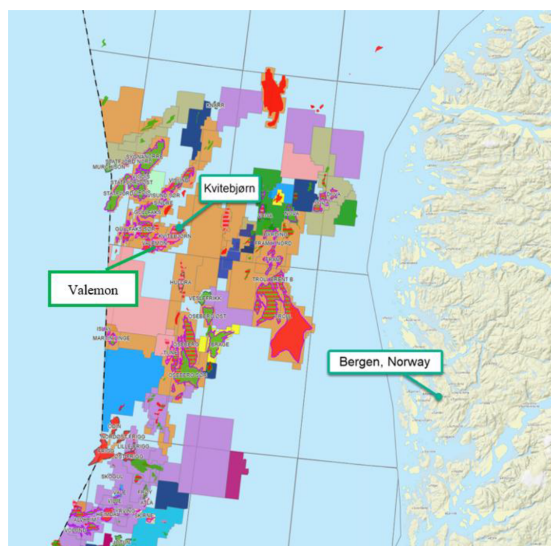


Figure 1—Kvitebjørn and Valemon Fields

Since production began in 2004, multiple drilling campaigns have been conducted to maintain output and access new areas of the field (Paulus, Murray, et al. 2023). As depletion progressed, new opportunities were identified in targets located farther from the platform. These flank targets were beyond the reach of the original well design envelope, requiring semi-extended reach drilling (ERD) with long overburden sections as illustrated in Figure 2 and Figure 3.

A-16 B Completion Schematic

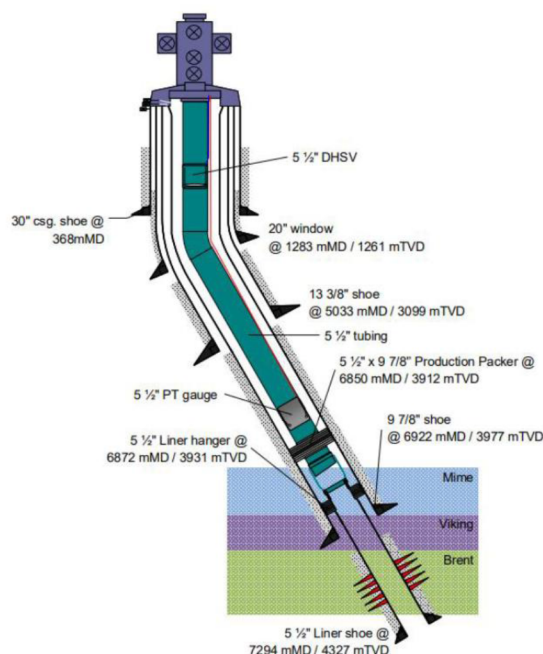


Figure 2—Kvitebjørn Well Design

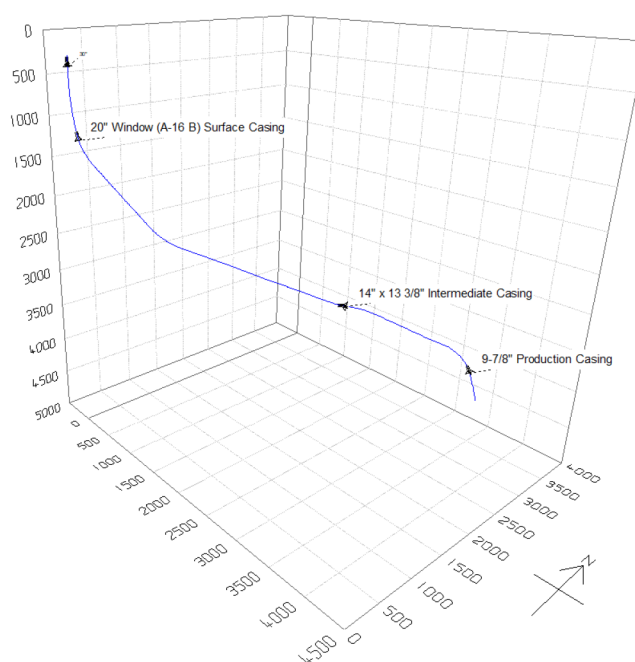


Figure 3—Kvitebjørn Wellpath

Valemon Field

The Valemon field which was discovered in 1985 and development sanctioned in 2011, is in the northern Norwegian sector of the North Sea as shown in Figure 1. Early planning classified the Brent and Statfjord reservoir targets as HPHT, with expected pressures of 797 bar at 155 °C (approx. 4,009 mTVDSS) for Brent and 865 bar at 170 °C (approx. 4,423 mTVDSS) for Statfjord (Paulus, Shen and Beeh, IADC/SPE-212448-MS *Becoming an HPHT World Class Project, the Valemon experience 2012-2017*. 2023).

Initial drilling was conducted with a jack-up rig prior to topside installation, due to the water depth of approximately 135 m. Following platform installation, subsequent wells were drilled from the fixed production facility. Similarly with Kvitebjørn, recent Valemon drilling campaigns targeted reservoir flanks, pushing well design beyond the platform's original drilling scope and requiring semi-ERD with long overburden intervals as illustrated in Figure 4 and Figure 5. These extended designs, coupled with complex formations, contributed to early campaign cost overruns and high non-productive time (NPT).

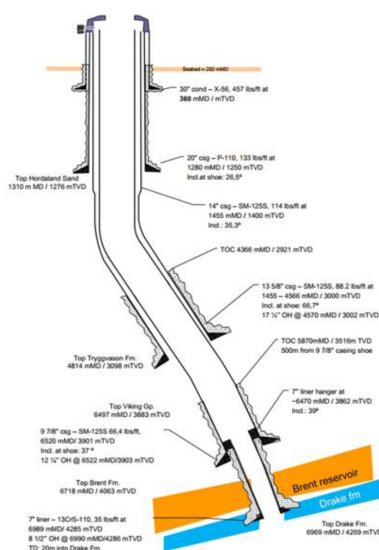


Figure 4—Valemon Well Design

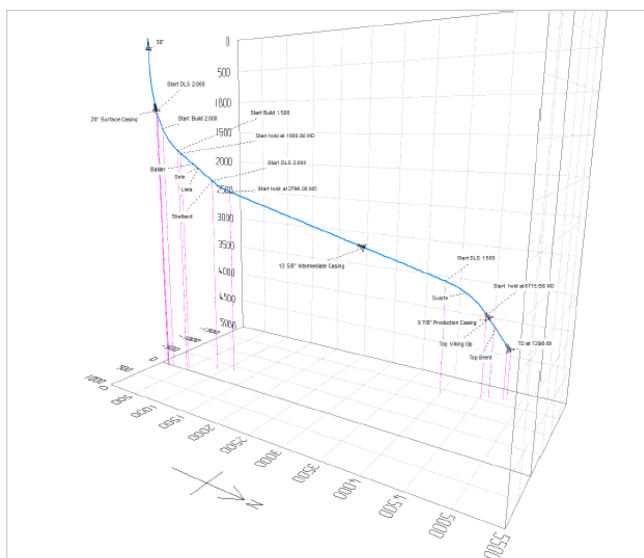


Figure 5—Valemon Wellpath

Wide Vision Drilling is a Mindset Shift

The Wide Vision Drilling is not just a framework or process; it is a mindset that should shift the way of thinking to ensure clear understanding of project specific challenges and innovate fit for purpose solutions. The two core principles of the Wide Vision Drilling are as follows:

- **From Technology-First to Understanding-First**

There is a growing trend in the industry to use advanced technologies as the first line of defense against complexity. In some cases, technology is applied without fully understanding the formation-specific risks, leading to false confidence, unanticipated failures, and expensive re-drilling.

Wide Vision Drilling challenges this practice. It starts with a simple principle: "Don't fix the risk—understand it, engineer it, and then design it out." This represents a philosophical departure from a technology-driven model to a risk-first engineering model. It doesn't reject tools, but insists they come after risk is understood, decomposed, and controlled through proper engineering design.

Engineers must return to their roots: comprehensive offset analysis, formation behavior, stress evolution, drilling mechanics, pressure trends, and wellbore stability physics. Only then should they consider what technology is required - if any.

- **From Speed to Appropriate Pace**

In modern drilling, performance is often misrepresented as drilling faster. Daily reports celebrate "meters per hour" or "connection times," reinforcing a culture that sees speed as success. But in HPHT wells where stress margins are tight and error windows narrow speed without control leads to increased risks, engineering/operational failures and eventually increased cost.

Wide Vision Drilling (WVD) challenges this mindset. It reframes performance not as raw speed, but as appropriate speed, a carefully engineered pace that:

- Matches the technical complexity of the formation,
- Anticipates risk evolution across each formation transition, and
- Prioritizes borehole quality and long-term integrity over short-term pace.

"In complex wells, performance is not defined by how fast you go, but by how far you get without failing."

WVD does not try to eliminate the natural tension between speed and safety, it embraces it as a design parameter. This roadmap walks the fine line between technical limits and operational risk, choosing the fastest possible speed that doesn't compromise stability, safety, or performance durability. This thinking flips the traditional performance curve on its head: instead of chasing maximum Rate of Penetration ROP, WVD targets the maximum sustainable performance envelope, unique to each formation group and well geometry

Wide Vision Drilling Framework

WVD is not just a work process, it is a framework for field-specific learning and risk transformation that is used throughout the well construction process. [Figure 6](#) illustrates the steps of the workflow. This paper will explain how the framework was used for successful delivery of the 17 ½-inch section in both Valemon and Kvitebjørn HPHT projects.

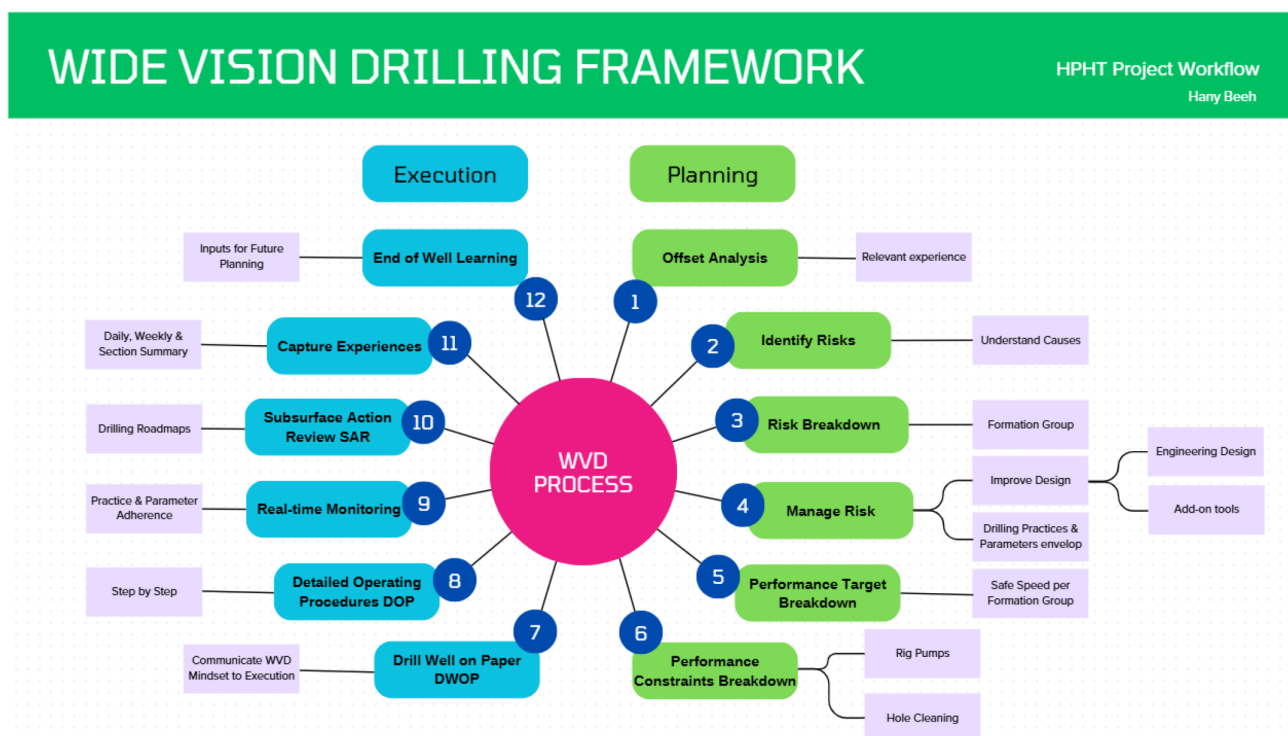


Figure 6—Wide Vision Drilling (WVD) Framework

Key pillars of the framework are split between planning phase and execution phase. The planning phase covers offset analysis, identification of risk and risk breakdown, managing the risk, performance target and performance constraints breakdown. The execution phase covers Drill Well on Paper workshop, Detailed Operational Procedures (DOP), real time monitoring and subsurface action review (SAR), capturing experience and end-of-well learning. These will be explained with examples in the following chapters and sub-chapters.

Planning Phase

Offset Analysis

In complex projects, it is essential to understand field-specific experiences, operational practices, and patterns of failures. Therefore, the first step in the Wide Vision Drilling (WVD) workflow was to conduct a comprehensive offset well analysis for every well drilled within the project since 2004 (Beeh, et al. 2025). This analysis involved a detailed review of daily drilling reports, end-of-well reports, experience reports, and incident records, complemented by the tacit knowledge of the project team members who were involved in the operations at the time.

The findings were presented through two visual tools:

- An event log illustrating subsurface risks by formation and depth as shown in Figure 7.
- Run summaries detailing the performance of individual operational phases as shown in Figure 8.

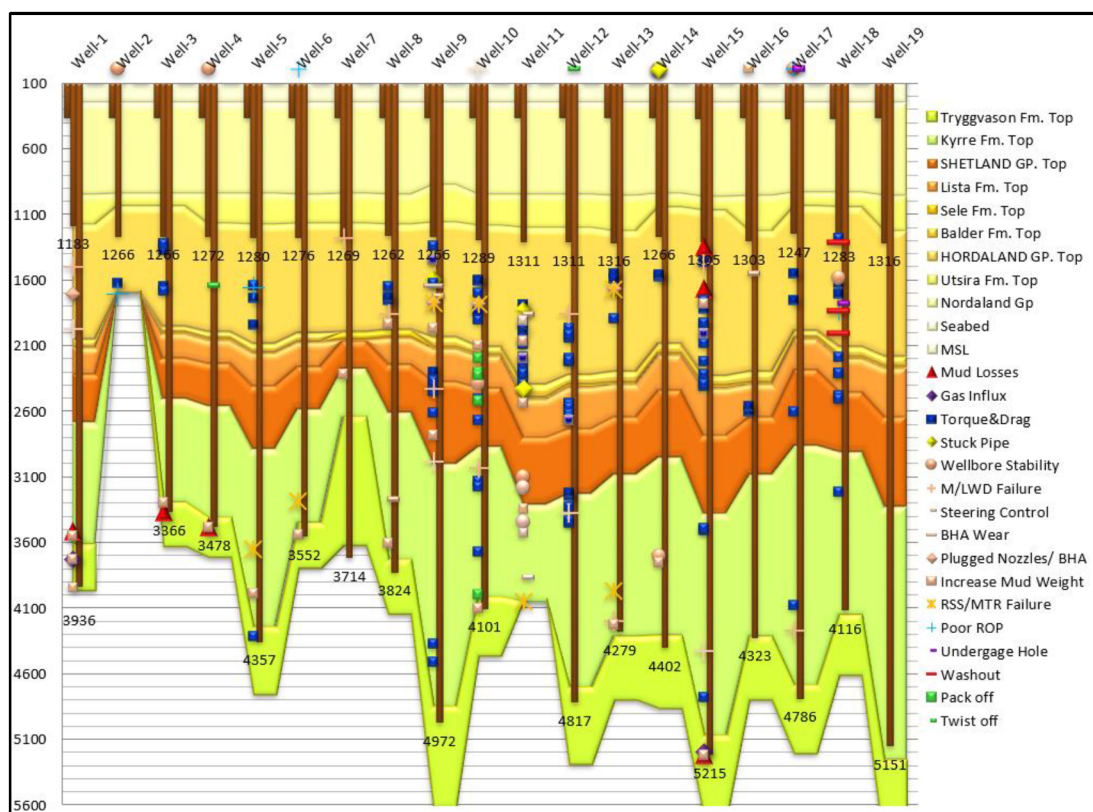


Figure 7—Event Log Plot for Offset Wells Study (Beeh, et al. 2025)

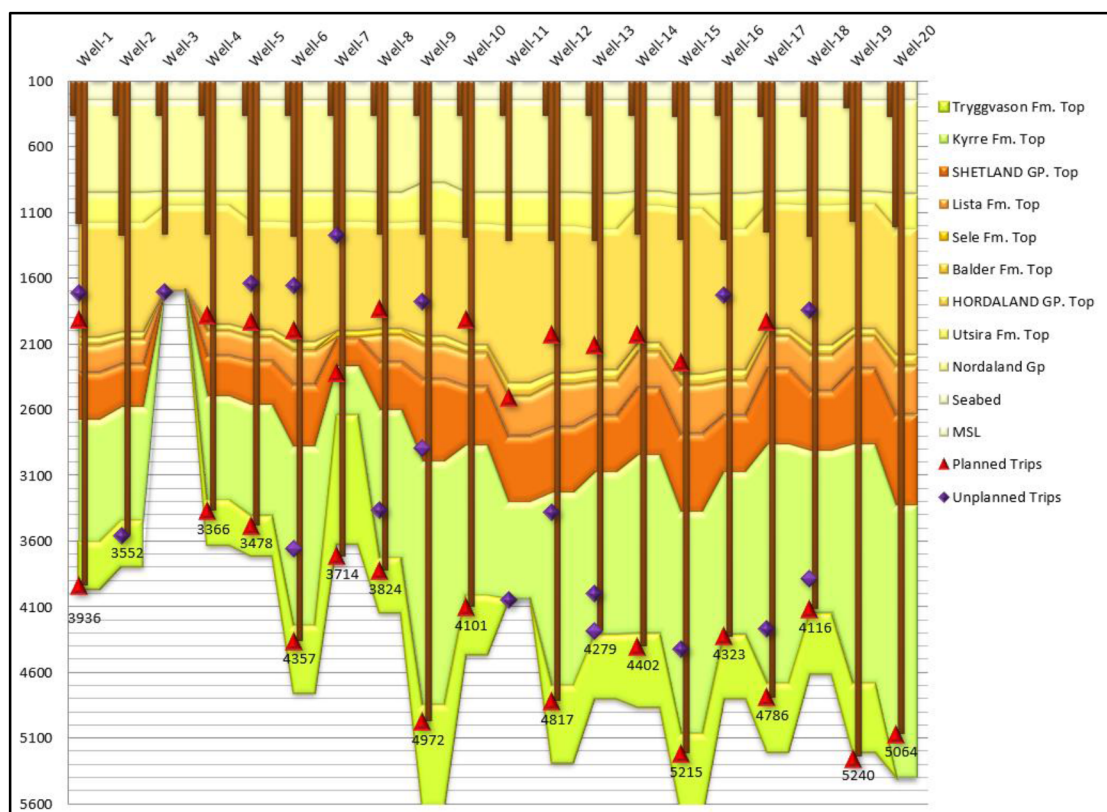


Figure 8—Run Summaries Plot for Offset Wells Study (Beeh, et al. 2025)

Identify Risks and Risk Breakdown

A key outcome of the offset analysis was to identify subsurface risks based on historical experiences and conduct engineering analysis to determine their possible causes.

The 17½-inch section in this HPHT project posed significant challenges due to its extended length, reaching up to 4,000 meters, and inclinations up to 70°. The well design became increasingly complex, with recent targets located in flank areas or within the reservoir. Meanwhile, the casing point for the 17½-inch section was maintained at a consistent depth between 3,000 and 3,200 meters True Vertical Depth (TVD) to provide the required formation strength for drilling the subsequent 12¼-inch section. Consequently, this resulted in a section length of approximately 4,000 meters with high inclination angles up to 70° (Beeh, et al. 2025).

Given the diverse lithologies encountered throughout this section, effective risk management required categorizing risks into three major formation groups:

- **Hordaland Group:**

- Hard Cemented Stringers: These caused significant issues such as damage to drill bits and Bottom Hole Assemblies (BHAs), tool failures, and reduced Rate of Penetration (ROP).
- Wellbore Instability and Formation Washouts: The unconsolidated sands and weak shales exhibited instability under excessive drilling parameters, resulting in washouts that impaired hole cleaning, caused loss of directional control, induced high vibrations, and complicated tripping and casing operations.
- Downhole Losses: Fluid losses were observed in Hordaland sands, necessitating precise mud weight management to mitigate risks.

- **Rogaland Group:**

- Fluid losses were observed in Hordaland sands, necessitating precise mud weight management to mitigate risks.

- **Shetland Group:**

- The Shetland formation group is more competent and stable compared to the Hordaland and Rogaland groups. However, drilling a deep, large section at high inclination in the avalanche zone posed hole cleaning challenges. These difficulties were compounded when the upper portion of the section included weaker formations (Hordaland and Rogaland Groups), requiring effective hole cleaning during drilling. Any attempt to backream out of the hole risked inducing instability in the upper section. Additionally, mud pumps occasionally delivered a maximum flow rate of 4,800 liters per minute (lpm), which resulted in relatively low annular velocities.

Manage Risks

With risks decomposed by formation groups, it became clear that each group presented specific subsurface challenges requiring tailored control measures. This rationale underpinned the decision to segment the section into formation groups or packages, ensuring that risk mitigation strategies were relevant and properly targeted for each group. Table 1 illustrates the corresponding control measures applied to each formation group.

Table 1—Planning, Execution Risk and Performance Target Breakdown (Beeh, et al. 2025)

Formation (Risk work-package)	Risks	Planning Control Measures	Execution Control Measures	Drilling Parameters
Hordaland Group	Tool damages due to drilling through cemented stringers	3-D durable bits cutting structure	Detailed stringers drilling procedures	Flowrate: 3800 - 4200 LPM
		Low inclination through stringers	Downhole Drilling Mechanics Tool in BHA	String Rotation: 100 - 140 RPM ROP 70 – 80 m/hr
	Formation washouts	Rotary Steerable System with wide flow range	Optimized drilling parameters	
		BHA optimized stabilization design	No off-bottom work	
Rogaland Group	Downhole losses	Designed mud weight strategy (start with low mud weight)	Close monitoring of ECD Contingency LCM flow chart	
	Formation Instability	Limit inclination to 50	Optimized drilling parameters	Flowrate: 3800 - 4500 LPM
		Increase mud weight to 1.46 SG	No off-bottom work	String Rotation: 140 - 160 RPM ROP 50 m/hr
Shetland Group	Hole cleaning	Design BHA for minimum pressure loss and maximize flowrate	Use maximum flow and string rotation	Flowrate: 4800 LPM
			Monitor hole cleaning roadmaps	String Rotation: 180 RPM
			Clean the hole while drilling	ROP 35 - 40 m/hr

Performance Target and Constraints Breakdown

Performance objectives were established in alignment with the subsurface risk profiles and technical limitations. The performance targets for the 17½-inch section were decomposed by formation groups, as summarized in Table 1, to balance risk levels with the technical constraints of rig equipment and downhole tools.

The primary goal is to walk the line between risk and performance to successfully complete the approximately four-kilometer section in one run, optimizing efficiency while minimizing failures and operational issues.

The focus was on delivering an efficient and safe section rather than rigidly adhering to micro-Key Performance Indicators (KPIs). Micro KPIs were used selectively, only when they supported the overall section delivery goal. In cases where micro KPIs conflicted with safe section delivery, priority was given to safety and successful section completion.

To clearly communicate this mindset across the integrated project team including the operator, rig contractor, and Main Service Provider (MSP) team, we developed a Well Charter as illustrated in Figure 9. This charter serves as a roadmap defining high-level section performance targets, guiding all efforts towards achieving these goals

	+ Initiatives	+ Happy	+ How?	+ Improvements
Whipstock	Evaluate WS design and Vendors. Manage 3rd part closely	Successful WS Exit	Manage 3rd party closely. Milling Roadmap	Evaluate elimination of Polish run
17,50 inch Section	Single Run in 17,5-in	Single run achieved, 3rd longest in KVB, 5th in EQN. Textbook drilling, flawless execution.	Offset well analysis. Local best practices, BHA design, mud control charts.	PowerDrive Stabilization.
	Increase ROP in all sections	Focus on total section delivery	Wide Vision Drilling (WVD), Optimized drilling	Low Inc., more ROP Cutting handling
	Remove S&V tool, LEAN, reduce risk & cost saving	Low S&V. Low MSE. Delivered section in One Run. Correct perception	Bit/ BHA design and Drilling practices	NA
	First two stage cement job	Successful primary and secondary cement job. Barriers achieved	Detailed planning. Caliper data for C-flex placement	Stage cementing collar preparation process. Customized cement fundament
12,25 inch Section	Single Run in 12,25-inch	Single run achieved, 2nd longest 12,25inch.	3-D Cutting structure bit, BHA design	MPD tuning and training. SLB can assist HAL with modeling and fluids inputs
	Increase ROP in all section	Doubled ROP in the 12,25 inch	3-D Cutting structure bit, BHA design, drilling roadmap	Jar placement Cuttings handling
8,50 inch Section	Single Run in 8,5-inch	Single run achieved for 1st time	DFMS, TST HDD kit and packer element. Implement RWP of previous incidents	New WSM strategy Testing and risk anlaysis
	Increase ROP in all sections	ROP was hindered by ECD spikes	Challenges of ECD spikes and overreaction of MPD system	Review Bit/BHA design (VAL experience) Deepen 9 7/8" casing

Figure 9—Well Charter for HPHT Project

Execution Phase

Drill Well on Paper DWOP Workshop

Drill Well on Paper (DWOP) is a critical pre-execution planning workshop where all key stakeholders (the operator, rig contractor, MSP, and service companies) collaborate to walk through the drilling plan step-by-step before actual operations commence.

DWOP has been a vital step within our framework to achieve the following objectives:

- Identify and eliminate potential risks, inefficiencies, and sources of non-productive time prior to mobilization, thereby ensuring safe, efficient, and cost-effective execution.
- Anticipate operational hazards and challenges.
- Develop contingency plans and preventive measures.

- Streamline workflows and align expectations concerning drilling sequence, logistics, and responsibilities.
- Identify opportunities to reduce flat-time and enhance operational efficiency.
- Ensure all parties fully understand the well design, execution plan, and their specific roles.
- Establish clear communication protocols.
- Incorporate lessons learned and field-specific drilling practices from offset wells or similar operations.

The ultimate objective of DWOP in the Wide Vision Drilling Framework is to communicate and embed the WVD mindset thoroughly with the execution team, ensuring it is embraced and followed during the execution phase. A sound plan is only as good as its execution. Therefore, DWOP is essential to officially hand over the plan and align the execution team on all aspects of the operation.

Detailed Operating Procedures

Detailed Operating Procedures (DOPs) are step-by-step written instructions that describe exactly how a specific operational task should be performed, including the sequence, responsibilities, safety checks, and acceptance criteria. DOPs are fundamental to operational readiness and risk management, ensuring all personnel perform tasks consistently, safely, and in compliance with engineering and HSE standards.

Given the length and complexity of the 17½-inch section, as it spans over multiple geological challenges, it was imperative to divide the section within the DOP into three distinct parts. Each part features its own drilling parameters, practices, and strategies tailored to the specific subsurface and drilling risks identified, as detailed in Figure 10. This approach provides the execution team with targeted guidance and clarity through each segment of the section.

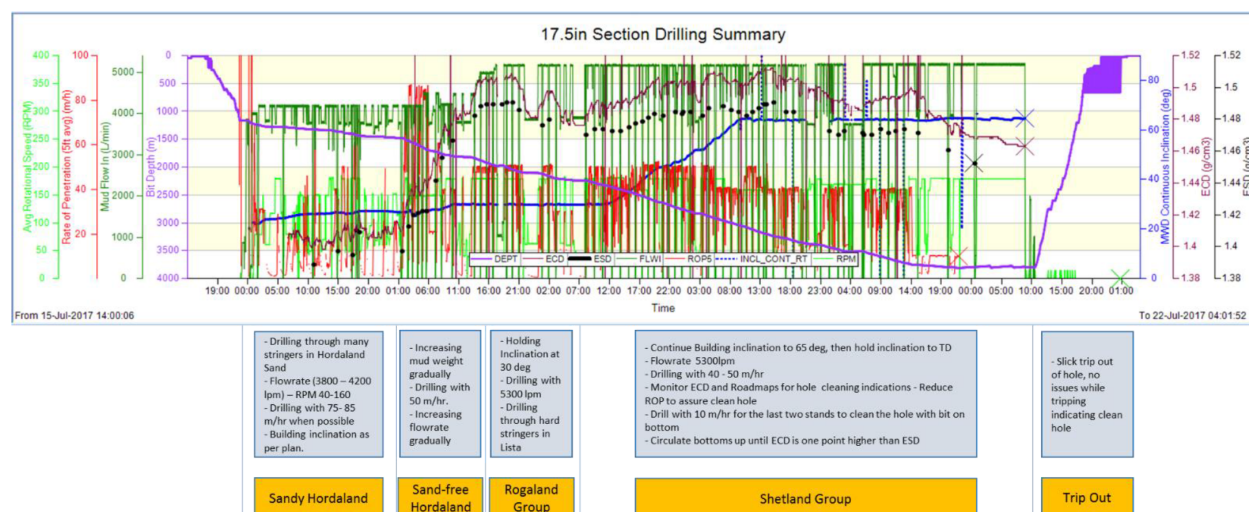


Figure 10—17 1/2-inch Section Drilling Summary (Beeh, et al. 2018)

Figure 10 summarizes the actual drilling for the 17½-inch section, visually presenting the modified drilling procedures and parameters used to complete the section in a single run (Beeh, et al. 2018). This figure is frequently referenced during the DWOP workshop to reinforce understanding and confidence in the framework among the execution team.

Real-time Monitoring and Subsurface Action Review (SAR)

Continuous monitoring of adherence to the agreed drilling roadmap, parameters, and practices outlined in the DOP is critical throughout well execution. It is equally important to monitor deviations or changes in real-time trends 24/7, enabling timely and informed responses. Understanding these deviations requires field-specific experience, which has proven indispensable in assessing changes accurately and initiating corrective actions that prevent secondary risks.

A structured communication platform, the Subsurface Action Review (SAR) meeting, is held twice daily between onshore and offshore teams. These sessions review drilling roadmaps and real-time sensor data to identify trend deviations, verify their significance, and collaboratively determine the optimal course of action for the next 12 hours, as illustrated in Figure 11.

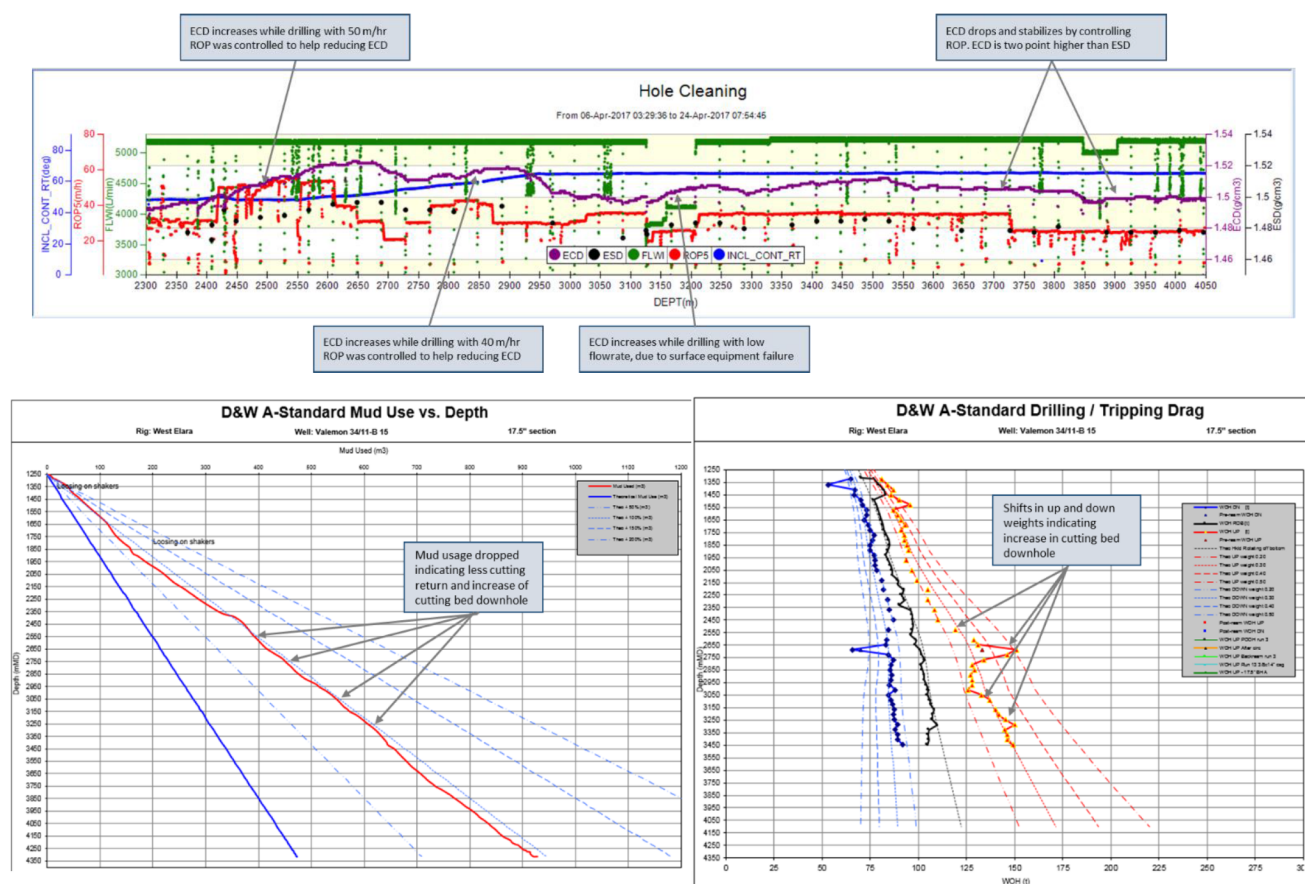


Figure 11—Realtime Sensors and Roadmap Trends Monitoring (Beeh, et al. 2018)

Capture Experience and End of Well Learning

The Wide Vision Drilling Framework places strong emphasis on knowledge capture and operational learning. Capturing experience and conducting formal end-of-well reviews are essential steps to transfer valuable operational knowledge across wells and campaigns, preventing repetition of past mistakes and encouraging replication of successful practices.

Lessons learned drive continuous improvement by integrating past insights into future well planning and execution. They enhance risk understanding and mitigation, helping avoid recurrence of incidents, equipment failures, or inefficiencies. Moreover, experience capture supports performance optimization by identifying and standardizing best practices across wells, rigs, and teams.

Figure 9 exemplifies this continuous learning mindset. Even though the performance targets outlined in the Well Charter were successfully met, the project team proactively documented lessons learned and identified opportunities for further improvement in future wells and drilling campaigns.

Wide Vision Drilling Framework Implementation Challenges

Implementing the Wide Vision Drilling Framework proved to be a complex endeavor. A well-designed process alone is insufficient without the full understanding, commitment, and adherence of the people involved. The journey faced several challenges from the outset and throughout the project lifecycle. Below are key obstacles encountered during the framework implementation:

- **Initial opposition to the framework:**

For years, drilling performance in HPHT projects had been primarily defined by maximizing speed. The common practice focused on increasing drilling rates without sufficiently considering the associated subsurface risks or how aggressive parameters could adversely affect well delivery. The Wide Vision Drilling Framework introduced the concept of variable and controlled penetration rates tailored to different formation groups. It also proposed modifications to drilling practices and the establishment of project-specific procedures that were previously uncommon. This approach initially met resistance from many established stakeholders as it does not fully align with operational KPIs. However, after presenting offset well analyses alongside a comprehensive risk management plan which included risk identification, understanding root causes, and applying control measures, the project team gradually embraced the framework. Further confidence increased following the successful drilling of the first well under the new methodology, which yielded unprecedented successful results.

- **Changes in One Team Personnel:**

The HPHT project had multiple drilling campaigns with intermittent pauses. Maintaining continuity in applying the framework was challenging as key personnel who championed the process were reassigned to other projects, and numerous new team members joined. Despite having a valuable learning database, much of the framework's success relied on tacit knowledge and critical insights passed down through experience which was not easily transferred to newcomers. Consequently, with the departure of the framework's original advocates and the influx of personnel accustomed to their own preferred practices from other projects, the framework's application weakened and was nearly forgotten. This emphasized the necessity to brand and formally document the framework, as done in this paper, to preserve institutional knowledge and prevent future dilution.

- **Chasing KPIs, and the Halo Effect (Authority Bias):**

At the start of a new drilling phase with mainly new team members, a strong performance-driven mindset led to an emphasis on speed as the key metric. During drilling of a long, highly inclined 17 1/2-inch section, this resulted in significant cuttings accumulation a known risk that the framework explicitly aimed to prevent through recommended parameters and drilling practices specific to formation groups. Despite clear mitigation guidelines, the team opted to ignore the framework's recommendations and followed general advice from external experts unfamiliar with the project-specific risk and processes. This decision led to subsurface failure and loss of the section, necessitating a costly re-drill that resulted in over a month of non-productive time as shown in Figure 14. This case exemplifies the halo effect or authority bias where decisions are disproportionately influenced by perceived authority figures without critical assessment, highlighting the importance of strict adherence to project-specific processes and the need for questioning and verification.

Summary, Results and Conclusion

Wide Vision Drilling WVM™ Effect on Section Delivery

The Wide Vision Drilling framework focuses on optimizing and maximizing the efficiency of engineering design and the drilling roadmap, while strictly maintaining safe operational limits. This workflow enabled significant improvements in well delivery and drilling efficiency on HPHT projects, especially in the challenging 17 ½-inch section.

In the Valemon Field, out of 22 drilled 17 ½-inch sections since 2011, the WVD workflow was applied in only 6 sections. Statistical analysis shows that the average drilling time for the 17 ½-inch section without WVD was 26.3 days. With WVD, the average was reduced to 12.9 days, effectively halving the section delivery time, as shown in Figure 12.

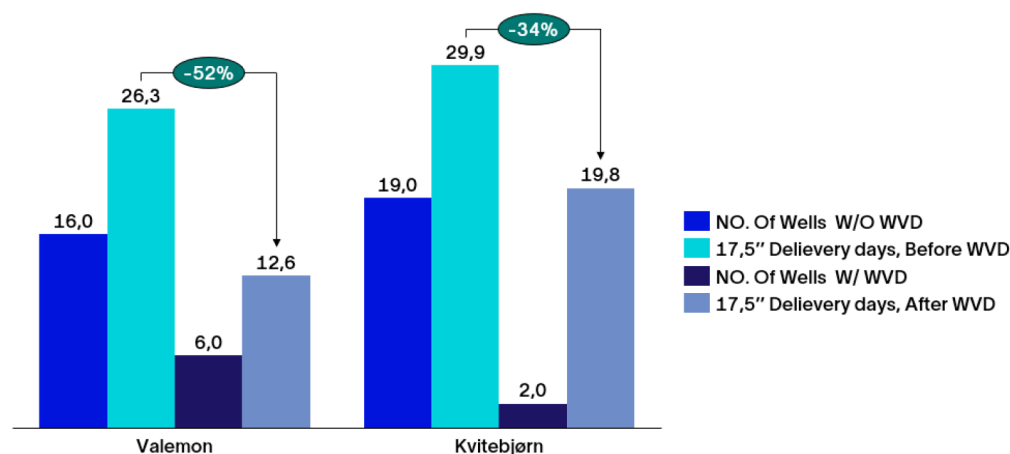


Figure 12—Wide Vision Drilling (WVD) Effect on Section Delivery

Similarly, on the Kvitebjørn Field, 21 of these sections have been drilled since 2004, with WVD applied in only 2 of them. Here, the average delivery time without WVD was 29.9 days, compared to 19.8 days with WVD, representing approximately a 34% reduction, as shown in Figure 12.

To illustrate this further, a comparison between two 17 ½-inch sections drilled on Kvitebjørn is presented: Well #1 drilled before WVD implementation, and Well #2 drilled using the WVD workflow. As shown in (Figure 13), although the 17 ½-inch section of Well #1 was drilled at a faster penetration rate, the total section delivery time for Well #2 was approximately four days shorter overall. This improved performance is attributed to:

1. Risk management during drilling: Optimizing drilling speed and following the roadmap effectively mitigated risks.
2. Positive impact on subsequent operations: The systematic WVD approach enabled smoother transitions and reduced downtime in follow-up operations.

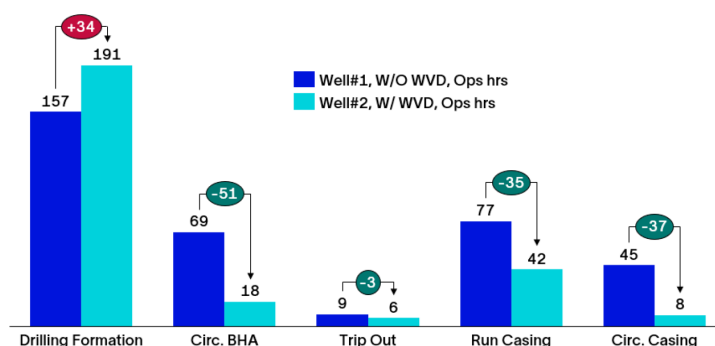


Figure 13—Wide Vision Drilling Improves Overall Delivery

The WVD workflow redefines project success by shifting the focus from pure speed toward balanced, optimized drilling performance.

Wide Vision Drilling WVM™ Effect on Non-Productive Time (NPT)

At the core of WVD lies deep risk understanding and control, aiming to proactively eliminate drilling risks. Strict adherence to the WVD practices and roadmap is crucial to minimizing subsurface challenges.

For example, in Valemon Field, Well #1 was drilled without WVD implementation. The team focused on achieving micro KPIs and maximizing drilling speed without a full understanding of field-specific risks or rig technical limits. The decision-makers were driven by competition with other projects that did not face similar subsurface challenges. Although the 17 ½-inch section was drilled faster than usual, at total depth circulation, cuttings were observed over the shaker. Despite this, the team decided to initiate a trip out with the Bottom Hole Assembly (BHA) in place. The trip quickly became problematic: the BHA became packed-off, downhole losses occurred, and eventually the drill string could not be recovered. This led to the decision to shoot off the string, set cement plugs, and sidetrack the well.

The sidetrack (Well #2) was drilled with WVD implementation. Following the setback in Well #1, the team was more cautious and drilled at a penetration rate slower than even the WVD recommendations. Despite the slower drilling pace, the section was delivered successfully.

Figure 14 compares the total section delivery times with and without WVD. In this case, WVD could have saved approximately 35 days of NPT if it had been applied during drilling of Well #1's 17 ½-inch section. The benefits of WVD increase with project complexity, where risk likelihood and impact are magnified. The figure also shows that although Well #1 drilled faster, the time saved was lost during difficult tripping operations to free the stuck drill string. Meanwhile, Well #2's section was approximately 4.5 days faster than Well #1 when summing formation drilling, circulation at total depth, and trip-out time.



Figure 14—Wide Vision Drilling Effect on Reducing Non-Productive Time and Improving Section Delivery

Key achievements from these projects are summarized as follows:

- 0 well control incidents
- 20% - 30% below wells budget time
- 50 Days of early production
- Significant reduction in Non-Productive Time
- 6 Wells successfully delivered below budget on Valemon.
- 2 Wells successfully delivered below budget on Kvitebjørn.
- 1 Run per section, for each 17 ½-inch section drilled with Wide Vision Drilling.
- 0 Downhole tool failures for each 17 ½-inch section drilled with Wide Vision Drilling.
- HPHT Project life extension due to wells budget reduction, and increased technical feasibility of drilling challenging wells

Conclusion

The WVD framework demonstrates that lasting operational excellence in complex HPHT wells depends on a disciplined balance between engineering precision, deep risk understanding, and adaptive execution. WVD redefined the approach to challenging wells by embedding a culture of continuous learning, anchoring practices in field-specific knowledge, and resisting the industry tendency to chase speed at the expense of. It's proven in both Kvitebjørn and Valemon project by delivering consistent performance despite personnel changes, positions it not only as a technical solution but also as a strategic enabler for long-term field development.

The methodology is not without its challenges and resistance. However, these organizational challenges are manageable if the team can understand project risks to re-define KPIs, onboard new personnel using the framework, and leadership that empowers the team to overcome the authority bias leading to poor decision making. The methodology sets a best-practice benchmark that can be applied to mature and complex fields worldwide, ensuring safer, more efficient, and economically optimized well delivery.

Acknowledgement

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The authors hoped that this publication would serve as a case study material or references for other relevant projects and provides development drilling opportunities in mature HPHT field with complex subsurface.

Abbreviations

BHA	– Bottom Hole Assembly
DOP	– Detailed Operational Procedures
DWOP	– Drill Well On Paper
ECD	– Equivalent Circulating Density
ERD	– Extended Reach Drilling
HPHT	– High Pressure High Temperature
KPI	– Key Performance Indicators

LCM	– Loss Circulation Materials
LPM	– Liters Per Minute
mMD	– meter Measured Depth
mTVD	– meter True Vertical Depth
mTVDSS	– meter True Vertical Depth Subsea
MSL	– Mean Sea Level
MSP	– Main Service Provider
NPT	– Non-Productive Time
PRV	– Pressure Relieve Valve
Psi	– pound per square inch
RKB	– Rotary Kelly Table
ROP	– Rate Of Penetration
RPM	– Rotation Per Minute
SAR	– Subsurface Action Review
Sg	– specific gravity
WVD	– Wild Vision Drilling

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